

Reflection Journal

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My main takeaway from this lab is that understanding data matters more than simply getting code to run. Pandas, NumPy, and matplotlib each help answer a different question: how the information is organized, what its numbers tell us, and what patterns are visible. I see the value of using these tools together because a table, a calculation, and a chart can explain different parts of the same dataset. This makes machine learning preparation feel more concrete: useful predictions begin with understanding the information available.

One distinction I need to keep clear is the difference between a feature and a label. The Iris table has five columns, but only four are measurement features. The species column contains the category that a future model would try to predict. Thinking of the table as a spreadsheet makes the structure easier to understand, but the purpose of each column matters more than its position. This distinction helps me connect ordinary data organization with the way machine learning problems are defined.

The numerical results also show why one summary is not enough. The average sepal length is about 5.84 cm, while the standard deviation is about 0.83 cm. I interpret the first number as the center of the measurements and the second as a measure of their spread. Neither number describes every flower. Comparing the averages by species makes the differences clearer, but even those averages cannot show all the variation within each group. This reminds me to ask what a calculation leaves out before drawing a conclusion.

The scatter plot provides the clearest example of that limitation. Setosa tends to form a relatively distinct group, while versicolor and virginica overlap considerably. I can see why a difference between group averages does not guarantee that individual flowers are easy to classify. The bar chart adds another perspective: all three species have 50 examples. Equal counts help describe the dataset, but they do not prove that a model would perform well. A useful next question is whether petal measurements would separate the species more clearly.

Another lesson I take from the notebook is that checking the work is part of the analysis. There are no missing values here, so filling missing data would be unnecessary. Running cells in order also matters because later steps depend on earlier variables. My approach to future labs should be to inspect the data, explain the outputs in plain language, and then check that the notebook runs from the beginning. Being able to explain why a result makes sense is a stronger sign of understanding than a success message alone.